

Here is your complete, fully-solved, and beautifully highlighted guide covering ONLY Course Outcome 1 (CO1) from your notes! 🚀 Let's dive into the core concepts and the numericals in a clean, visual format. ✨

🎲 📖 COMPLETE FUN GUIDE TO PROBABILITY & RANDOM VARIABLES (CO1) ✨

☀️ UNIT 1 — PROBABILITY BASICS

🎲 What is Probability?

Probability measures the chance that an event will occur. 👉 If something is impossible → probability = 0 👉 If something is certain → probability = 1 Mathematical Bound: $0 \leq P(E) \leq 1$

🎯 Random Experiment & Sample Space

Random Experiment: A process whose result cannot be predicted exactly beforehand is called a Random Experiment. ✅ **Examples:** Throwing a dice 🎲, Tossing a coin 🪙, Drawing a card 🃏. Even if conditions are same, result may change.

Outcome: A possible result of a random experiment.

Sample Space (SS): The set of all possible outcomes is called the Sample Space. *Example:* Throwing a dice gives possible outcomes $S = \{1, 2, 3, 4, 5, 6\}$. Each number is an outcome.

🧩 Types of Events

Any collection of outcomes from the sample space is called an Event.

- 1 **Elementary Event:** Contains only one outcome. (Example: Getting "4" on a dice, $E = \{4\}$)
- 2 **Compound Event:** Contains more than one outcome. (Example: Getting an even number, $E = \{2, 4, 6\}$)
- 3 **Impossible Event** ❌: Event that can never occur ($P(E) = 0$).
- 4 **Certain Event** ✅: Event that always occurs ($P(E) = 1$).
- 5 **Complementary Event** 🔄: If A occurs, A' does not occur. $P(A') = 1 - P(A)$
- 6 **Mutually Exclusive Events** 🚫: Two events that cannot occur together ($A \cap B = \emptyset$).
- 7 **Exhaustive Events** 🌐: Events whose union gives the entire sample space ($A_1 \cup A_2 \cup \dots = S$).
- 8 **Equally Likely Events** ⚖️: Events having the same chance of occurring.

🎯 Classical & Axiomatic Probability

Classical Probability Formula: If total outcomes = n and Favourable outcomes = m , then: $P(E) = \frac{m}{n}$

Axiomatic Laws:

- ♦ **First Law:** $P(E) \geq 0$ (Probability is never negative).
- ♦ **Second Law:** $P(S) = 1$ (Total probability of sample space is 1).
- ♦ **Third Law (Addition Law):** If mutually exclusive: $P(A \cup B) = P(A) + P(B)$ If NOT mutually exclusive: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

☀️ CONDITIONAL PROBABILITY & BAYES' THEOREM

🔗 Independent vs. Dependent Events

Independent Events: Occurrence of one event does not affect another (e.g., Two coin tosses). **Dependent Events:** Occurrence of one event affects another (e.g., Drawing cards without replacement).

Conditional Probability

Probability of A given B already occurred. $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Bayes' Theorem

Used to reverse conditional probabilities. $P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum P(E_i)P(A|E_i)}$

RANDOM VARIABLES & DISTRIBUTIONS

What is a Random Variable?

A variable whose value depends on outcomes of a random experiment. Usually denoted by X, Y, Z . 1

Discrete Random Variable: Takes finite/countable values. (Examples: Dice number, Number of heads). 2

Continuous Random Variable: Takes infinite values over an interval. (Examples: Height, Weight, Time).

Distribution Functions

PMF (Probability Mass Function): Used for discrete random variables. $f(x) = P(X=x)$ Properties:  $f(x) \geq 0$  $\sum f(x) = 1$

PDF (Probability Density Function): Used for continuous variables. Properties:  $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x) dx = 1$

CDF (Cumulative Distribution Function): $F(x) = P(X \leq x)$ *Important Relation:* For a continuous variable: $F(x) = \int_{-\infty}^x f(t) dt$ and $f(x) = \frac{d}{dx}F(x)$

EXPECTATION, VARIANCE & STANDARD DEVIATION

Expectation / Mean ($E(X)$): Average value of a random variable. *For discrete:* $E(X) = \sum xf(x)$ *For continuous:* $E(X) = \int xf(x) dx$

Variance ($Var(X)$): Measures the spread of data. *Formula:* $Var(X) = E(X^2) - [E(X)]^2$

Standard Deviation (σ): *Formula:* $\sigma = \sqrt{Var(X)}$

Important Properties:  Variance of Constant: $Var(c) = 0$  Variance Scaling: $Var(aX+b) = a^2Var(X)$

FULLY SOLVED CO1 NUMERICALS

Basic Probability & Independent Events

1. Dice Total of 7

- Question:** Two dice are thrown simultaneously. Find the probability of getting a total score of 7.

- **Solution:** Since 2 dice are used, the sample space has 36 possible outcomes. The favorable outcomes that sum to 7 are (1,6), (2,5), (3,4), (4,3), (5,2), and (6,1). The required probability is $6/36 = 1/6$.

2. Seating Arrangements

- **Question:** 6 boys and 6 girls sit in a row randomly. Find the probability that: (i) all 6 girls sit together, and (ii) boys and girls sit alternately.
- **Solution (i):** Consider the 6 girls together as 1 object. This gives 7 objects to arrange in 7! ways. The 6 girls can arrange themselves internally in 6! ways. Total ways = 12!. Probability is $\frac{7! \times 6!}{12!} = \mathbf{\frac{1}{132}}$.
- **Solution (ii):** For alternating seats, there are 2 patterns (BGBG... or GBGB...). The boys can be arranged in 6! ways and the girls in 6! ways. Probability is $\frac{2 \times 6! \times 6!}{12!} = \mathbf{\frac{1}{462}}$.

3. Math Problem Solved by Students

- **Question:** A math problem is given to 3 students. Their chances of solving it are $1/2$, $1/3$, and $1/4$. Find the probability that the problem is solved.
- **Solution:** The probability the problem is solved equals 1 minus the probability that none of them solve it. Probability none succeed = $(1 - 1/2) \times (1 - 1/3) \times (1 - 1/4) = 1/2 \times 2/3 \times 3/4 = 1/4$. Probability solved = $1 - 1/4 = \mathbf{\frac{3}{4}}$.

Conditional Probability & Bayes' Theorem

4. The Two Bags Problem

- **Question:** Bag 1 contains 6 Black balls, and Bag 2 contains 4 Black and 3 Black balls (as written in notes). 1 ball is drawn at random and found to be black. Find the probability it was drawn from Bag 1.
- **Solution:** Let E_1 be the event of choosing Bag 1 and E_2 be Bag 2. The probability of choosing either is $P(E_1) = P(E_2) = 1/2$. Probability of drawing Black from Bag 1: $P(A|E_1) = 6/10$. Probability of drawing Black from Bag 2: $P(A|E_2) = 3/7$. Using Bayes' Theorem: $P(E_1|A) = \frac{P(A|E_1) \times P(E_1)}{P(A|E_1) \times P(E_1) + P(A|E_2) \times P(E_2)}$

5. The Three Urns Problem

- **Question:** Urn I contains 1 White, 2 Black, 3 Red balls. Urn II contains 2 White, 1 Black, 1 Red. Urn III contains 4 White, 5 Black, 3 Red. One urn is chosen at random and 2 balls are drawn. They are White and Red. Find the probability it is Urn II.
- **Solution:** The probability of picking any urn is $P(E_1) = P(E_2) = P(E_3) = 1/3$. Probability of drawing White and Red from Urn I: $P(A|E_1) = \frac{1C_1 \times 3C_1}{6C_2}$. Probability of drawing White and Red from Urn II: $P(A|E_2) = \frac{2C_1 \times 1C_1}{4C_2}$. Probability of drawing White and Red from Urn III: $P(A|E_3) = \frac{4C_1 \times 3C_1}{12C_2}$. Applying Bayes' Theorem yields a final probability of 58/118.



Probability Density Functions (Continuous)

6. Continuous PDF Evaluation

- **Question:** If a random variable X has a PDF $f(x) = 1/4$ on the interval $[-2, 2]$, obtain $P(2x+3 > 5)$.

- **Solution:** Simplify the inequality: $2x+3 > 5$ implies $x > 1$. $P(x > 1) = \int_1^2 \frac{1}{4} dx = \mathbf{\frac{1}{4}}$

7. Finding PDF Constants & CDF

- **Question:** A continuous random variable has the PDF $y = K(x-1)(2-x)$ for $1 \leq x \leq 2$. Determine K , the CDF, and $P(5/4 \leq x \leq 3/2)$.
- **Solution:** To find K , integrate the PDF over the interval and set it equal to 1: $\int_1^2 K(3x - x^2 - 2) dx = 1$ implies $\mathbf{K = 6}$. The CDF $F(x)$ is found by integrating the PDF from 1 to x : $F(x) = \mathbf{5 - 12x + 9x^2 - 2x^3}$ (as written in notes) $P(5/4 \leq x \leq 3/2) = F(3/2) - F(5/4) = \mathbf{\frac{11}{32}}$.

★ Expectation, Variance & PMF (Discrete)

8. Finding 'c', Expectation, and Variance

- **Question:** A PMF is given as $f(0) = c$, $f(1) = 2c - 3c^2$, and $f(2) = 4c - 1$. Find c , the conditional probability $P(x \geq 1 | x < 2)$, and the Expectation and Variance of X .
- **Solution:** The sum of probabilities must equal 1: $c + 2c - 3c^2 + 4c - 1 = 1$. This simplifies to $3c^2 - 7c + 2 = 0$, giving $c = 2$ or $c = 1/3$. Since probabilities cannot exceed 1, $\mathbf{c = 1/3}$. $P(x \geq 1 | x < 2) = \frac{P(x=1)}{P(x=0) + P(x=1)} = \frac{1/3}{1/3 + 1/3} = \mathbf{\frac{1}{2}}$. Expectation $E(X) = 0(1/3) + 1(1/3) + 2(1/3) = \mathbf{1}$. To find Variance, first find $E(X^2) = 0^2(1/3) + 1^2(1/3) + 2^2(1/3) = \frac{5}{3}$. Variance $= E(X^2) - [E(X)]^2 = 5/3 - 1^2 = \mathbf{\frac{2}{3}}$.